

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

WTW158 Calculus

Semester Test 2

11 May 2024

Time (all sections): 135 minutes

Marks: 50

M-NUMBER:

SURNAME:	[REDACTED]
FIRST NAMES:	[REDACTED]
STUDENT NUMBER:	[REDACTED]
SIGNATURE:	[REDACTED]

Lecture group. Indicate in which group you are:

Group 1. Dr Tegegn	Group 2. Prof Kufakunesu	Group 3. Dr Emereuwa	Group 4. Dr Jooste	Group 5. Dr Dumani
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INSTRUCTIONS

- The question paper consists of pages 1 to 16. Check that there are no pages missing.
- Rough working can be done on the blank pages available for this purpose. No rough working will be marked, however.
- No electronic devices may be used.
- Answers for multiple choice SECTION A must be circled on this question paper, then entered with a soft pencil on Side 2 of the optical mark reader form. No answers may be erased from the optical mark reader form. The optical mark reader form must be handed in 1 hour after the start of the test.
- Answers for SECTION B and SECTION C must be written in blue or black ink on the question paper and in the space provided (graphs must also be in ink).
- Should you need more space for an answer in Section C, you may use any blank page. Indicate clearly where you will complete the answer. On the page where you complete the answer, you also need to indicate clearly which part completes the answer.
- Work done in pencil or in red ink will not be marked.
- The use of correction fluid ("Tipp-Ex") is prohibited.
- Any queries about the marking must be raised within three days after the marks have been released. You will also be able to see your marked script at that stage.

MARKS Sections B and C	
B (13)	9 ✓
p.10-11 (8)	5½ ✓
p.12-13 (7)	5.5 ✓
p.14-15 (6)	0.5 ✓
Total (34)	20½ ✓

SECTION A [Multiple choice questions] - 16 marks

QUESTION 1

Consider $t(x) = e^{2x}$. Then $t'(\ln 3) =$

(a) 6	(b) 9	(c) 12	(d) 15	(e) 18	(f) 24	(g) None of (a) - (f)
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$$t(x) = e^{2x} \quad t'(x) = 2e^{2x} \quad 2e^{2(\ln 3)} = 2e^{\ln 3^2} = 2e^{\ln 9} = 2 \cdot 9 = 18$$

QUESTION 2

If $m'(x) = \arctan(x) + \frac{x}{1+x^2}$, then $m''(1) =$

(a) $\frac{\pi}{4}$	(b) $\frac{\pi}{4} + \frac{1}{2}$	(c) $\frac{3}{2}$	(d) $\frac{1}{2}$	(e) $\tan 1 + \frac{1}{2}$	(f) None of (a) - (e)
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$$m''(x) = \frac{1}{1+x^2} + \frac{x \left(\frac{d}{dx}(1+x^2) \right) - (1+x^2)}{(1+x^2)^2} = \frac{1}{1+x^2} + \frac{x(2x) - (1+x^2)}{(1+x^2)^2}$$

QUESTION 3

The y -intercept of the tangent line to the curve with equation $y = \sqrt{x} - x$ at the point $(1, 0)$ is

(a) $-\frac{1}{2}$	(b) $\frac{1}{2}$	(c) $\frac{1}{4}$	(d) $-\frac{1}{4}$	(e) None of (a) - (d)
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$$\begin{aligned} m''(1) &= \frac{1}{1+1^2} + \frac{1(2(1)) - (1+1^2)}{(1+1^2)^2} \\ &= \frac{1}{2} + \frac{2-2}{2^2} = \frac{1}{2} + \frac{0}{4} = \frac{1}{2} \end{aligned}$$

QUESTION 4

Consider the following statements:

- (1) If $f(x) = \ln(3x^3)$, then $f'(x) = \frac{3}{x}$. ✓
- (2) If $f(x) = \sinh(3x)$, then $\frac{f''(x)}{f(x)} = -9$. ✗
- (3) If $f(x) = e^{3x}$, then $f'(x) + f'(-x) = 3(f(x) + f(-x))$. ✓

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

QUESTION 5

 $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x + 1} - x) =$

(a) 0	(b) $\frac{1}{2}$	(c) 1	(d) $\sqrt{3} - 1$	(e) $\sqrt{2} - 1$	(f) Limit does not exist
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$$\begin{aligned} \lim_{x \rightarrow \infty} (\sqrt{x^2 + 2x + 1} - x) &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2 + 2x + 1} - x)(\sqrt{x^2 + 2x + 1} + x)}{\sqrt{x^2 + 2x + 1} + x} \\ &= \lim_{x \rightarrow \infty} \frac{x^2 + 2x + 1 - x^2}{\sqrt{x^2 + 2x + 1} + x} = \lim_{x \rightarrow \infty} \frac{2x + 1}{\sqrt{x^2 + 2x + 1} + x} \\ &= \lim_{x \rightarrow \infty} \frac{2 + \frac{1}{x}}{\sqrt{1 + \frac{2}{x} + \frac{1}{x^2}} + 1} = \frac{2}{\sqrt{1 + 0 + 0} + 1} = \frac{2}{2} = 1 \end{aligned}$$

QUESTION 6

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be any function. Consider the statements:

- (1) If f is continuous at 3 then f is differentiable at 3. ✗
 (2) If f is not differentiable at 3 then f is not continuous at 3. ✗
 (3) If f is not continuous at 3 then f is not differentiable at 3. ✓
 (4) If f is differentiable at 3 then f is continuous at 3. ✓

Which of the above statements are necessarily true? Choose the correct answer:

- (a) Statements (1), (2), (3) and (4) are true;
 (b) Only statements (2), (3) and (4) are true;
 (c) Only statements (3) and (4) are true;
 (d) Only statements (2) and (4) are true;
 (e) Only statements (1), (3) and (4) are true.
 (f) None of (a) - (e)

QUESTION 7

Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$. Consider the following statements:

- (1) If $\lim_{x \rightarrow 5} g(x) = 0$ then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist. ✗
 (2) If $\lim_{x \rightarrow 3} f(x) = 0$ then $\lim_{x \rightarrow 3} \frac{1}{f(x)} = \infty$. ✓
 (3) If $\lim_{x \rightarrow 1} f(x) = \infty$ then $\lim_{x \rightarrow 1} \frac{9}{f(x)} = 0$. ✓

Which of the above statements are necessarily true?

- (a) Only statement (1);
 (b) Only statement (2);
 (c) Only statement (3);
 (d) Only statements (1) and (2);
 (e) Only statements (1) and (3);
 (f) Only statements (2) and (3);
 (g) Statements (1), (2) and (3);
 (h) Statements (1), (2) and (3) are false.

QUESTION 8

If $g(x) = \tan^{-1}(x) = \arctan(x)$, then $\lim_{x \rightarrow 1} \frac{g(x) - g(1)}{x - 1} =$

(a) 0	(b) $\frac{1}{2}$	(c) 1	(d) 2	(e) $\frac{\pi}{4}$	(f) None of these.
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End of Section A

$$\frac{1}{1+x^2} = \frac{1}{2}$$

$$\lim_{x \rightarrow 1} \frac{\arctan(x) - \arctan(1)}{x - 1} = \frac{\frac{1}{1+x^2} - \frac{1}{1+1}}{1} = \frac{\frac{1}{2} - \frac{1}{2}}{1} = \frac{0}{1} = 0$$

SECTION B [Fill in questions] - 13 marks

SECTION B: Write only your final answer in the block provided. Only this will be marked. Rough working can be done outside the blocks and will not be marked.

QUESTION 9 [5 marks]

$$f'(x) = \frac{1}{\sqrt{1-x^2}} \cdot 2x$$

Find the derivatives of the following. Do not simplify your answers.

(9.1) $f(x) = \arcsin(x^2)$

$$f'(x) = \frac{1}{\sqrt{1-x^2}} (2x) = \frac{2x}{\sqrt{1-x^2}} \quad \times$$

(9.2) $f(x) = \frac{\ln(x)}{x}$

$$f(x) = \frac{d}{dx} \ln(x) x^{-1} = \frac{1}{x} (x^{-1}) + \ln(x) (-x^{-2}) = \frac{1}{x^2} - \frac{\ln x}{x^2} = \frac{1 - \ln x}{x^2}$$

$$f'(x) = \frac{1 - \ln(x)}{x^2} \quad \checkmark \quad (2)$$

(9.3) $f(x) = \tan(\tan(\tan(x)))$

$$\frac{d}{dx} \tan(\tan(\tan(x))) = \sec^2(\tan(\tan(x))) \cdot \frac{d}{dx} \tan(\tan(x))$$

$$f'(x) = \sec^2(\tan(\tan(x))) \cdot \sec^2(\tan(x)) \cdot \sec^2(x) \quad \checkmark \quad (2)$$

$$= \sec^2(\tan(\tan(x))) \cdot \sec^2(\tan(x)) \cdot \sec^2(x) \cdot 1$$

QUESTION 10 [2]

Let f be a function with $f'(x) = 1 - \frac{1}{x-2}$. On which interval(s) is f decreasing?

$$\text{for } x \neq 2 \text{ and } f \text{ is always increasing} \quad \times$$

$$f'(x) = 1 - \frac{1}{x-2} \quad f'(x) < 0 \text{ if } f \text{ is decreasing}$$

$$\therefore f'(x) < 0 \text{ if } x > 3 \quad 1 - \frac{1}{-3-2} = 1 - \frac{1}{-5} = 1 + \frac{1}{5} > 0$$

QUESTION 11 [2]

Determine the critical numbers of the function $f(x) = \sin(x) + \cos(x)$ on the interval $[0, 2\pi]$.

$$\frac{\pi}{4}, \frac{5\pi}{4} \quad \checkmark \quad (2)$$

$$f(x) = \sin(x) + \cos(x)$$

$$\therefore f'(x) = \cos(x) - \sin(x)$$

$$\cos x - \sin x = 0$$

$$\therefore \cos x = \sin x$$

$$\therefore x =$$

0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	
0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	

$$1 - \frac{1}{-1-2} = 1 + \frac{1}{3} > 0$$

$$1 - \frac{1}{-3-2} = 1 + \frac{1}{5} > 0$$

QUESTION 12 [2]

Write down the statement (conditions and result) of the Mean Value Theorem.

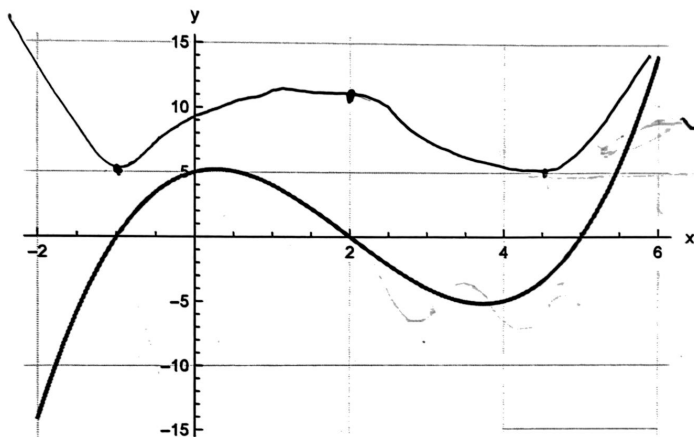
for a function f on the domain (a, b)
 f must be continuous on $[a, b]$
 f must be differentiable on (a, b)
 $\therefore$ for a value c , $\frac{f(b) - f(a)}{b - a} = f'(c)$

②

QUESTION 13 [2 marks]

(a) The graph of h' is given. Sketch the graph of h on the same set of axes, given that $h(-1) = 5$.

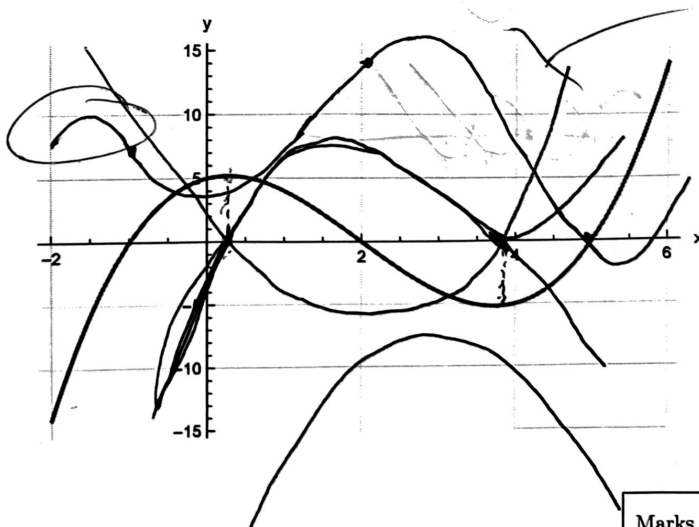
(1)



①

(b) The graph of h' is given. Sketch the graph of h'' on the same set of axes.

(1)



①

Marks Section B:

9

if $h'' < 0$ graph is concave down
 if $h'' > 0$ graph is concave up

End of Section B

SECTION C [Full written answers] - 21 marks

QUESTION 14 [2 marks]

Find the derivative of $y = x^{\sin(x)}$.

$$y = x^{\sin(x)}$$

$$\therefore \ln y = \ln(x)^{\sin(x)}$$

$$\therefore \ln y = \sin(x) \ln(x)$$

$$\therefore \frac{d}{dx}(\ln y) = \frac{d}{dx}[\sin(x) \ln(x)]$$

$$\therefore \frac{1}{y} \frac{dy}{dx} = \frac{d}{dx}[\sin(x) \ln(x) + \sin(x) \frac{d}{dx}(\ln(x))]$$

$$\therefore \frac{1}{y} \frac{dy}{dx} = \cos x \ln(x) + \sin x \cdot \frac{1}{x}$$

$$\therefore \frac{dy}{dx} = y \left[\cos x \ln(x) + \frac{\sin x}{x} \right]$$

$$y = x^{\sin(x)} \therefore y' = x^{\sin(x)} \left[\cos x \ln(x) + \frac{\sin x}{x} \right]$$

QUESTION 15 [2 marks]

Determine the limit, if it exists. Show all your steps.

$$\lim_{x \rightarrow 1^+} \left(\frac{2}{x^2 - 1} - \frac{1}{x - 1} \right)$$

$$\lim_{x \rightarrow 1^+} \left(\frac{2}{x^2 - 1} - \frac{1}{x - 1} \right) = \frac{\infty}{\infty} - \frac{1}{0} \text{ indeterminate form.}$$

$$\therefore \lim_{x \rightarrow 1^+} \left(\frac{2}{x^2 - 1} - \frac{1}{x - 1} \right) = \lim_{x \rightarrow 1^+} \left[\frac{2x - 2 - 1(x^2 - 1)}{(x^2 - 1)(x - 1)} \right]$$

$$= \lim_{x \rightarrow 1^+} \left[\frac{2x - 2 - x^2 + 1}{(x^2 - 1)(x - 1)} \right]$$

$$= \lim_{x \rightarrow 1^+} \left[\frac{-x^2 + 2x - 1}{(x^2 - 1)(x - 1)} \right]$$

$$= \lim_{x \rightarrow 1^+} \left[\frac{-(x - 1)(x - 1)}{(x^2 - 1)(x - 1)} \right]$$

$$= \lim_{x \rightarrow 1^+} \left(\frac{1 - x}{x^2 - 1} \right)$$

$$= \lim_{x \rightarrow 1^+} \left(-\frac{(x - 1)}{(x - 1)(x + 1)} \right)$$

$$= \lim_{x \rightarrow 1^+} -\frac{1}{x + 1}$$

$$= -\frac{1}{1 + 1} = -\frac{1}{2}$$

$$\therefore \lim_{x \rightarrow 1^+} \left(\frac{2}{x^2 - 1} - \frac{1}{x - 1} \right) = -\frac{1}{2}$$

QUESTION 16 [4 marks]

Find the horizontal asymptotes of the function

$$h(x) = \begin{cases} \frac{x}{\sqrt{3x^2+1}} & \text{if } x < -4 \\ \frac{x}{x+7} & \text{if } x > 0. \end{cases}$$

$$h(x) = \begin{cases} \frac{x}{\sqrt{3x^2+1}} & ; x < -4 \\ \frac{x}{x+7} & ; x > 0 \end{cases}$$

horizontal asymptotes: but find limit at ∞ :

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{3x^2+1}}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2(3 + \frac{1}{x^2})}}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{x\sqrt{3 + \frac{1}{x^2}}}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{\sqrt{3 + \frac{1}{x^2}}}$$

$$= \lim_{x \rightarrow \infty} (1)$$

$$\lim_{x \rightarrow \infty} (\sqrt{3 + \frac{1}{x^2}})$$

$$= \frac{1}{\lim_{x \rightarrow \infty} (3 + \frac{1}{x^2})}$$

$$= \frac{1}{\sqrt{\lim_{x \rightarrow \infty} 3 + \lim_{x \rightarrow \infty} \frac{1}{x^2}}}$$

$$\lim_{x \rightarrow \infty} \frac{1}{x^2} \Rightarrow 0$$

$$\therefore \lim_{x \rightarrow \infty} \frac{x}{\sqrt{3x^2+1}} = \frac{1}{\sqrt{3+0}} = \frac{1}{\sqrt{3}}$$

$$\lim_{x \rightarrow \infty} \frac{x}{x+7} = \lim_{x \rightarrow \infty} \frac{x}{x(1 + \frac{7}{x})}$$

$$= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{7}{x}}$$

$$= \frac{\lim_{x \rightarrow \infty} (1)}{\lim_{x \rightarrow \infty} (1 + \frac{7}{x})}$$

$$\lim_{x \rightarrow \infty} (1 + \frac{7}{x})$$

$$= \frac{1}{\lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{7}{x}}$$

$$= \frac{1}{1+0}$$

$$= \frac{1}{1}$$

$$\therefore \lim_{x \rightarrow \infty} \frac{x}{x+7} = 1$$

$\therefore$ the ~~has~~ has horizontal asymptotes at $\frac{1}{\sqrt{3}}$ and 1

$\therefore \frac{1}{\sqrt{3}} = h(x)$ has horizontal asymptotes at $\frac{1}{\sqrt{3}}$ and 1

QUESTION 17 [3 marks]

Consider the function $h: \mathbb{R} \rightarrow \mathbb{R}$, defined by $h(x) = \begin{cases} x^2 \sin \frac{1}{x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

Show that h is continuous at $x = 0$.

if h is continuous at 0,

$$\lim_{x \rightarrow 0} h(x) = h(0) \quad \text{if } \lim_{x \rightarrow 0} h(x) = h(0)$$

$$\therefore \lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} \left[x^2 \sin \left(\frac{1}{x^2} \right) \right] = \lim_{x \rightarrow 0} h(x)$$

$$\therefore \lim_{x \rightarrow 0} h(x) = \lim_{x \rightarrow 0} \left(x^2 \sin \left(\frac{1}{x^2} \right) \right)$$

(2.5)

by squeeze theorem,

$$-1 \leq \sin(x) \leq 1 \quad \checkmark, \quad x \in \mathbb{R} \setminus \{0\}$$

$$\therefore -1 \leq \sin \left(\frac{1}{x^2} \right) \leq 1$$

$$\therefore -x^2 \leq x^2 \sin \left(\frac{1}{x^2} \right) \leq x^2$$

$$\therefore \lim_{x \rightarrow 0} (-x^2) \leq \lim_{x \rightarrow 0} \left[x^2 \sin \left(\frac{1}{x^2} \right) \right] \leq \lim_{x \rightarrow 0} (x^2) \quad \checkmark$$

$$\therefore 0 \leq \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x^2} \leq 0$$

write separately

$$\therefore \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x^2} = 0$$

for $h(0)$:

$$h(0) = 0$$

$$\therefore \lim_{x \rightarrow 0} h(x) = h(0) = 0$$

$\therefore h$ is continuous at $x = 0$

QUESTION 18 [4 marks]

The equation

$$xy - 2 = \ln(x^2 + y - 4)$$

defines $y = f(x)$ implicitly as a differentiable function of x . Use implicit differentiation to find $f'(2)$ if $f(2) = 1$.

using implicit differentiation:

$$\frac{d}{dx} y = \frac{d}{dx} f(x) \Rightarrow \frac{dy}{dx} = f'(x)$$

(3)

$$\frac{d}{dx}(xy) - \frac{d}{dx}(2) = \frac{d}{dx}[\ln(x^2 + y - 4)]$$

$$\therefore \frac{d}{dx}(x) \cdot y + x \frac{dy}{dx} - 0 = \frac{1}{x^2 + y - 4} \cdot \frac{d}{dx}[x^2 + y - 4]$$

$$\therefore y + x \frac{dy}{dx} = \frac{1}{x^2 + y - 4} \cdot \left(\frac{d}{dx}(x^2) + \frac{dy}{dx} - \frac{d}{dx}(4) \right)$$

$$\therefore y + x \frac{dy}{dx} = \frac{1}{x^2 + y - 4} \cdot \left(2x + \frac{dy}{dx} - 0 \right)$$

$$\therefore y + x \frac{dy}{dx} = \frac{2x}{x^2 + y - 4} + \frac{1}{x^2 + y - 4} \cdot \frac{dy}{dx}$$

$$\text{if } y = f(x) \text{ then } xy - 2 = \ln(x^2 + y - 4) \quad \therefore y + x \frac{dy}{dx} - \frac{1}{x^2 + y - 4} \cdot \frac{dy}{dx} = \frac{2x}{x^2 + y - 4} - y$$

$$\therefore f(x) = \ln(x^2 + y - 4)$$

$$\therefore \frac{dy}{dx} \left[x - \frac{1}{x^2 + y - 4} \right] = \frac{2x}{x^2 + y - 4} - y$$

$$\therefore f(2) = 1$$

$$\therefore \ln(x^2 + y - 4) = 1$$

$$\therefore \frac{dy}{dx} = \frac{\left(\frac{2x}{x^2 + y - 4} - y \right)}{x - \frac{1}{x^2 + y - 4}} = f'(x)$$

$$f(x) = \frac{d}{dx} [x^2 + y - 4]$$

using implicit differentiation:

$$\therefore \frac{dy}{dx}$$

$$f(x) = y \quad \ln(x^2 + y - 4) = xy - 2$$

$$\therefore f'(2) = \frac{2(2) - y}{x^2 + y - 4} = \frac{2 - 1}{2^2 + y - 4}$$

$$\text{if } f(2) = 1 \quad \therefore \ln(2^2 + y - 4) = 2(2) - 2$$

$$\therefore \ln(8 + y) = 4 - 2 = 2$$

$$\therefore \ln(8) \cdot \ln y = 2 \cdot 2 = 4$$

$$f(x) = \frac{d}{dx}$$

$$\frac{d}{dx} f(x) = \frac{d}{dx} [\ln(x^2 + y - 4)]$$

$$\therefore f'(x) = \frac{1}{x^2 + y - 4}$$

$$= \frac{4 - y}{8 + y} = \frac{4 - y}{8 + y} \cdot \frac{8 + y}{16 + 2y - 1}$$

$$2 - \frac{1}{8 + y} = \frac{4 - y}{16 + 2y - 1}$$

Marks p.12, 13:

5.5

QUESTION 20 [3 marks]

Suppose $f(0) = 4$ and $f'(x) \leq 5$ for all values of x . How large can $f(2)$ possibly be? Show and motivate all your steps.

$$\text{if } f(0) = 4, f'(x) \leq 5 \quad \text{for } x \in \mathbb{R}, f(2) = ?$$

$$\text{if } f(0) = 4 \text{ and } f'(x) \leq 5$$

$$\therefore f(x) \leq 5x + 4$$

Why?

using $y = mx + c$, where $y = f(x)$, $m = f'(x)$
and $c = f(0)$

$$\therefore \text{find for } f(2):$$

$$f(2) \leq 5(2) + 4$$

$$\therefore f(2) \leq 10 + 4$$

$$\therefore f(2) \leq 14$$

$\therefore$ the largest possible value of $f(2)$ is 14

Marks p.14, 15:

0.5

End of Section C